

Original Article

Determination of Optimal Vapor Pressure Data by the Second and Third Law Methods

Kunihisa Nakajima

Nuclear Science and Engineering Center, Japan Atomic Energy Agency, 2-4 Shirakata, Tokai-mura, Naka-gun, Ibaraki 319-1195, Japan

Though equilibrium vapor pressures are utilized to determine thermodynamic properties of not only gaseous species but also condensed phases, the obtained data often disagree by a factor of 100 and more. A new data analysis method is proposed using the so-called second and third law procedures to improve accuracy of vapor pressure measurements. It was found from examination of vapor pressures of cesium metaborate and silver that the analysis of the difference between the second and third law values can result in determination of an optimal data set. Since the new thermodynamic method does not require special techniques and or experiences in dealing with measured data, it is reliable and versatile to improve the accuracy of vapor pressure evaluation.



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INTRODUCTION

Equilibrium vapor pressure data are utilized to determine thermodynamic properties of not only gaseous species but also condensed phases. But, vapor pressure measurement generally has a large uncertainty as mentioned in the preface of a handbook on thermodynamic data,¹⁾ “vapor pressures are not infrequently reported which disagree by a factor of 100 or more,” or as said in another paper,²⁾ “experience in vapor pressure measurements, particularly at high temperatures, has shown that large systematic errors are common, even among experienced investigators.” Moreover, another thermodynamic data hand book³⁾ says “when calorimetric data were available, they were given considerably more weight than values determined from vapor pressure data.” Based on this situation, the so-called second and third law procedures are considered as a method to improve accuracy of vapor pressure measurement. The second and third law methods^{4,5)} are commonly used to determine reaction enthalpy from equilibrium constant measured in experiment, for example, to determine vaporization enthalpy in terms of vapor pressure measurement. In addition, examination of the difference of the determined enthalpies between those calculated by the second and third law methods has been utilized to check whether the Gibbs energy functions and equilibrium data are mutually consistent⁴⁾ or to check whether cause of systematic errors are under control.⁵⁾ As for Gibbs energy functions, even if in the absence of experimental data, satisfactory estimates can usually be made by

a variety of procedures and uncertainties calculated from experimental molecular parameters are said to be small, *i.e.*, $\sim \pm 0.2$ to $0.4 \text{ J K}^{-1} \text{ mol}^{-1}$.⁵⁾ These uncertainties could correspond to a two to five percent error in the equilibrium constant which is assumed to have the same uncertainty level as the Gibbs energy functions.

Therefore, such a thermodynamic method based on the second and third laws can be used to improve accuracy of vapor pressure data especially if Gibbs energy functions are determined based on experimental data. This paper concerns about application of the second and third law methods for determination of optimal vapor pressure data. The new evaluation method is practically applied for equilibrium vapor pressure data of cesium metaborate, CsBO_2 ,⁶⁾ which is one of the most important chemical species having significant radiological impact in a severe accident like the one that occurred at Fukushima Daiichi Nuclear Power Station. Vapor pressures of silver given by nine laboratories⁷⁾ were also evaluated by the present method.

THERMODYNAMIC METHOD FOR DETERMINATION OF OPTIMAL VAPOR PRESSURES

The new thermodynamic method to determine optimal vapor pressures proposed in this study is based on the second and third law methods. In the third law method,^{8,9)} enthalpy changes of investigated reactions, $\Delta_r H^\circ$, can be determined from each experimental value of equilibrium constant, K_p° , using the following relation:

Correspondence to: Kunihisa Nakajima, Nuclear Science and Engineering Center, Japan Atomic Energy Agency, 2-4 Shirakata, Tokai-mura, Naka-gun, Ibaraki 319-1195, Japan, e-mail: nakajima.kunihisa@jaea.go.jp

$$\Delta_r H_{3rd}^\circ(\theta) = -RT \ln K_p^\circ(T) - T \Delta_r \text{gef}^\circ(T) \quad (1)$$

where θ is reference temperature of 0 or 298.15 K, R gas constant, T absolute temperature, and gef Gibbs energy function. Δ signifies the difference between thermodynamic quantities of products and reactants. The subscript (r) denotes reaction. The superscript (°) refers to standard-state pressure of 1 bar. The Gibbs energy function is defined as

$$\text{gef}^\circ(T) = -S^\circ(T) + (H^\circ(T) - H^\circ(\theta))/T \quad (2)$$

where S is entropy. Gibbs energy function of gaseous species can be calculated from molecular parameters by the use of statistical mechanics. While, Gibbs energy function of condensed phases can be obtained from heat capacity C_p° from 0 K to the temperature of interest. Equation (1) is derived through the relation:

$$\begin{aligned} -RT \ln K_p^\circ(T) &= \Delta_r G^\circ(T) \\ &= \Delta_r H^\circ(T) - T \Delta_r S^\circ(T) \\ &= T(\Delta_r H^\circ(T) - \Delta_r H^\circ(\theta))/T + \Delta_r H^\circ(\theta) - T \Delta_r S^\circ(T) \\ &= T \Delta_r \text{gef}^\circ(T) + \Delta_r H^\circ(\theta) \end{aligned} \quad (3)$$

On the other hand, in the second law method,¹⁰⁾ van't Hoff equation:

$$\Delta_r H^\circ(T)/T^2 = R d(\ln K_p^\circ(T))/dT \quad (4)$$

or

$$\Delta_r H^\circ(T) = -R d(\ln K_p^\circ(T))/d(1/T) \quad (5)$$

is used. Equation (5) is derived by substituting $dT = -T^2 d(1/T)$ into Eq. (4) or from the relation:

$$\begin{aligned} -R d(\ln K_p^\circ)/d(1/T) &= d(\Delta_r G^\circ(T)/T)/d(1/T) \\ &= d(\Delta_r H^\circ(T)/T)/d(1/T) - d(\Delta_r S^\circ(T))/d(1/T) \\ &= \Delta_r H^\circ(T) \end{aligned} \quad (6)$$

assuming that both of $\Delta_r H^\circ(T)$ and $\Delta_r S^\circ(T)$ are constant with temperature. Thus, $\Delta_r H^\circ(T_{\text{ave}})$ at average temperature, T_{ave} , is obtained from measured $K_p^\circ(T)$ by a least-squares analysis of an Arrhenius plot:

$$R \ln K_p^\circ(T) = A - \Delta_r H^\circ(T_{\text{ave}})/T \quad (7)$$

where both of A and $\Delta_r H^\circ(T_{\text{ave}})$ are fitting parameters. It is noted that T_{ave} is determined from the average value of $1/T$ for the measurements. Then, $\Delta_r H^\circ(T_{\text{ave}})$ is reduced to the reference temperature θ K through the use of heat content functions, $\Delta_r H^\circ(T_{\text{ave}}) - \Delta_r H_{2nd}^\circ(\theta)$:

$$\Delta_r H_{2nd}^\circ(\theta) = \Delta_r H^\circ(T_{\text{ave}}) - (\Delta_r H^\circ(T_{\text{ave}}) - \Delta_r H^\circ(\theta)) \quad (8)$$

But this procedure does not take into account for the temperature dependence of $\Delta_r H^\circ(T)$ within the measured temperature range. So more precise $\Delta_r H_{2nd}^\circ(\theta)$ can be obtained from the so-called Σ plot^{11,12)}:

$$\begin{aligned} R \ln K_p^\circ(T) - \left[\int_0^T \left(\frac{\Delta_r C_p^\circ dT''}{T'^2} \right) dT' + \frac{\Delta_r H^\circ(\theta) - \Delta_r H^\circ(0)}{T} \right] \\ = A' - \Delta_r H_{2nd}^\circ(\theta)/T \end{aligned} \quad (9)$$

where T' and T'' are variables of the integrations and A' is a fitting parameter. This equation is obtained by integrating the van't Hoff equation, Eq. (4), so that a Σ plot of measured

$K_p^\circ(T)$ will have linear dependence on $1/T$.

Now, this integration can further be changed to the following equation:

$$\int_0^T \left(\frac{\int_0^{T'} \Delta_r C_p^\circ dT''}{T'^2} \right) dT' = - \frac{\int_0^T \Delta_r C_p^\circ dT}{T} + \int_0^T \frac{\Delta_r C_p^\circ}{T} dT \quad (10)$$

Then, Eq. (9) can be transformed to the following equation:

$$\begin{aligned} A' - \Delta_r H_{2nd}^\circ(\theta)/T \\ = R \ln K_p^\circ(T) - \left[-(\Delta_r H^\circ(T) - \Delta_r H^\circ(0))/T + \Delta_r S^\circ(T) \right. \\ \left. - \Delta_r S^\circ(0) + (\Delta_r H^\circ(\theta) - \Delta_r H^\circ(0))/T \right] \\ = R \ln K_p^\circ(T) + \Delta_r \text{gef}^\circ(T) + \Delta_r S^\circ(0) \end{aligned} \quad (11)$$

and, when temperature-independent constants of A' and $\Delta_r S^\circ(0)$ are combined into a single constant A'' for simplicity, similar equation to Eq. (1) can be obtained:

$$-R \ln K_p^\circ(T) - \Delta_r \text{gef}^\circ(T) = A'' + \Delta_r H_{2nd}^\circ(\theta)/T \quad (12)$$

Actually, this equation can also be derived by integrating the van't Hoff equation, Eq. (4), from 0 K to the temperature of interest and we can find the following equation:

$$-R \ln K_p^\circ(T) - \Delta_r \text{gef}^\circ(T) = \Delta_r S^\circ(0) + \Delta_r H_{2nd}^\circ(\theta)/T \quad (13)$$

by use of the relation:

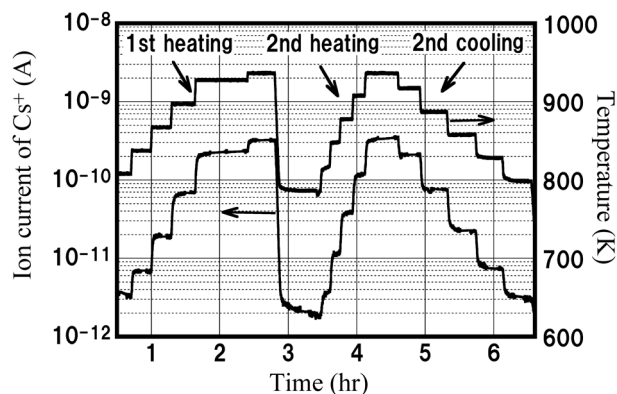
$$-RT \ln K_p^\circ(T)(T \rightarrow 0 \text{ K}) = \Delta_r G^\circ(0) = \Delta_r H^\circ(0)$$

Thus, in the case of the second-law method, two or more measured $K_p^\circ(T)$ are necessary to determine the value of $\Delta H_{2nd}^\circ(\theta)$ since the second law of thermodynamics cannot provide the relation: $\Delta_r S^\circ(0)=0$.¹³⁾ On the other hand, the third law value can be obtained even from a single measured $K_p^\circ(T)$ by use of Eq. (1) since the third law of thermodynamics can provide $\Delta_r S^\circ(0)=0$. In addition, when the temperature is multiplied on both sides of Eqs. (12) or (13), the left-hand side is the same as the right-hand side of Eq. (1). Then, if both the measured $K_p^\circ(T)$ and $\Delta_r \text{gef}^\circ(T)$ are accurate, A'' can become close to zero and $\Delta H_{2nd}^\circ(\theta)$ can also agree well with $\Delta H_{3rd}^\circ(\theta)$. Therefore, examination of difference of the values between $\Delta H_{2nd}^\circ(\theta)$ and $\Delta H_{3rd}^\circ(\theta)$ can result in determination of optimal data set of vapor pressures. In some cases, application of such a thermodynamic method might lead to choice of wrong data set of vapor pressures due to fortuitous compensations of errors between Gibbs energy function and vapor pressure data. However, since vapor pressure data are said to have a large uncertainty and accuracy of Gibbs energy functions is inferred to be commonly higher than that of vapor pressure data, such compensations are unlikely to happen. In other words, even if such compensations happen, vapor pressure data are considered to be the same level of errors as Gibbs energy functions.

APPLICATION TO VAPOR PRESSURE MEASUREMENT OF CESIUM METABORATE AND SILVER

Vapor pressure of CsBO₂

The detailed procedures of the sample and the Knudsen effusion mass spectrometric, KEMS, measurement

Fig. 1. Time and temperature dependences of the ion currents of Cs⁺.Table 1. The second-law and third-law sublimation enthalpies of CsBO₂.

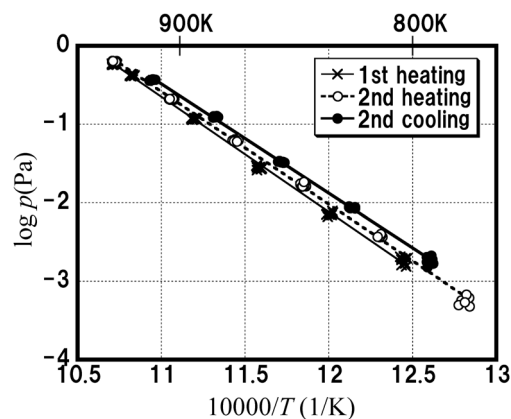
Authors	$\Delta_{\text{sub}}H_{2\text{nd}}^{\circ}(298)$ (kJ/mol)	$\Delta_{\text{sub}}H_{3\text{rd}}^{\circ}(298)$ (kJ/mol)	Diff. (%)
Nakajima <i>et al.</i> ⁽⁶⁾ (1st heating) for CsBO ₂ (s)	292.8	283.2	3.4
Nakajima <i>et al.</i> ⁽⁶⁾ (2nd heating) for CsBO ₂ (s)	287.1	282.0	1.8
Nakajima <i>et al.</i> ⁽⁶⁾ (2nd cooling) for CsBO ₂ (s)	278.4	279.9	0.6
Biswas and Mukerji ⁽¹⁸⁾ for CsBO ₂ (l)	237.6	284.3	16.4
Cordfunke <i>et al.</i> ⁽¹⁹⁾ for CsBO ₂ (l)	286.5	270.2	6.0
Cordfunke <i>et al.</i> ⁽¹⁹⁾ for CsBO ₂ (s)	289.1	271.9	6.3

are described in another paper.⁽⁶⁾ Briefly speaking, CsBO₂ sample used for the KEMS measurement was prepared from Cs₂CO₃ and H₃BO₃ mixed powders by heating at 873 K in air. In the KEMS measurement, the sample was heated and cooled stepwise in 30 K steps and the heating process was repeated two times. According to the paper⁽¹⁴⁾ which examined an ionization behavior of CsBO₂ by a KEMS method, detected Cs⁺ ion results from the dissociative ionization of CsBO₂ molecule. So only the ion current of Cs⁺ ionized under 50 eV electron impact, I_{Cs} , was monitored until the end of the KEMS measurement, as shown in Fig. 1. The measured ion current was converted into absolute vapor pressure by using the following equation⁽⁶⁾:

$$p_{\text{CsBO}_2}(t) = \frac{\sqrt{2\pi} R \Delta W_{\text{CsBO}_2} I_{\text{Cs}}(t) T(t)}{a L \int_0^{t_{\text{end}}} I_{\text{Cs}}(t') \sqrt{T(t')} dt' \sum_i \gamma_i \sqrt{M_i}} \quad (14)$$

where ΔW_{CsBO_2} is the weight of total amount of CsBO₂ effusing through the orifice, R the gas constant, a cross section of effusion orifice, L the Clausing factor,⁽¹⁵⁾ T absolute temperature, i an isotopic species, M the mass number of CsBO₂, γ isotopic abundance ratio of CsBO₂. ΔW_{CsBO_2} is assumed to be the same as the weight difference of the samples before and after the KEMS measurement. Ion currents used in conversion into absolute vapor pressures were chosen for the last one minute before the next heating or cooling step.

Figure 2 shows the determined vapor pressures plotted in Arrhenius form. As shown in this graph, there is a certain difference of vapor pressures between the heating and cooling processes. The ion currents at each temperature

Fig. 2. Temperature dependence of the vapor pressures of CsBO₂.

ultimately seem to be independent of time from Fig. 1. So it is hard from the time dependence of the ion current to determine which data set of vapor pressures is more accurate. Therefore, sublimation enthalpies of CsBO₂ by the second and third law methods are examined to find optimal data set of the vapor pressures. The Gibbs energy function of gaseous CsBO₂ required in this method is derived from molecular parameters given by Ezhov and Komarov⁽¹⁶⁾ assuming a rigid rotor harmonic oscillator model. However, it is found that there is a negligible difference between the Gibbs energy functions calculated from other molecular parameters reported in the past.⁽⁶⁾ While, the Gibbs energy function of solid CsBO₂ is computed from its standard entropy of formation and heat capacity recommended by Cordfunke and Konings,⁽¹⁷⁾ which are regarded as good quality. Therefore, accuracy level of both of Gibbs energy functions of solid and gaseous CsBO₂ are expected to be better than that of vapor pressures of CsBO₂. Table 1 shows examination results of the second-law and third-law sublimation enthalpies. As shown in the table, the difference between the second-law and third-law values for the 2nd cooling stage of Nakajima *et al.* is the smallest. Indeed, Fig. 2 indicates the vapor pressures increase in the order of the 1st heating, 2nd heating and 2nd cooling stages. Further, when we take a closer look at Fig. 1, the ion currents at the heating stages, especially at the 2nd heating stage, still seem to increase with time and then the equilibrium might not be reached yet. Consequently, since the present thermodynamic method requires neither special technique nor experience when dealing with measured data, it can be useful to determine an optimal data set of vapor pressures.

Table 1 also includes the second-law and third-law sublimation enthalpies evaluated from vapor pressures of CsBO₂ reported in the past.^(18,19) Further, the values derived from the left-hand side of Eq. (12) and from the right-hand side of Eq. (1) are plotted in Figs. 3 and 4, respectively. Figure 3 also includes regression lines extrapolated to near the origin. As shown in these figures, the data of our work is the most reasonable because the intercept of the regression line is the nearest of the origin and temperature dependence of the third law values is the weakest. Thus, the second and third law methods can also allow us to find a more preferable data set among literature data.

Vapor pressure of Ag

Vapor pressure measurements of silver were conducted

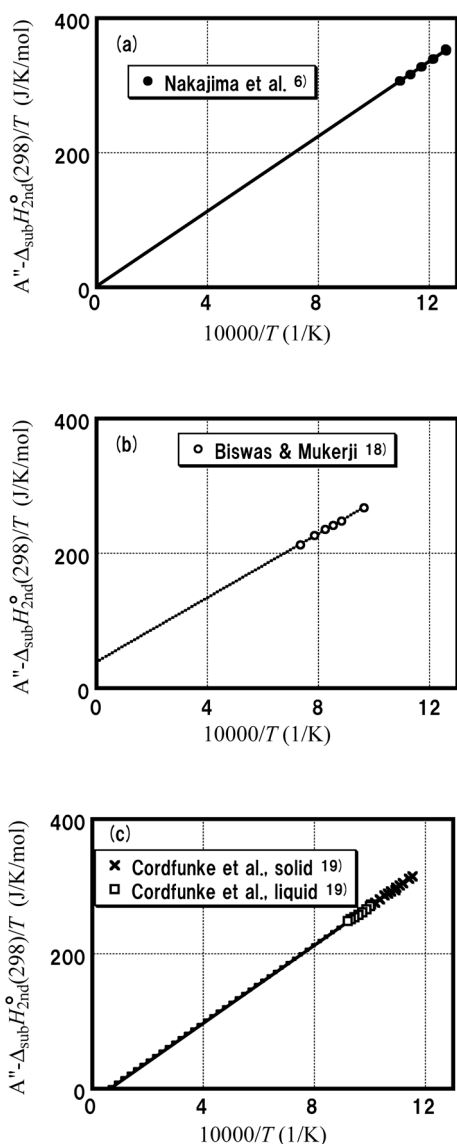


Fig. 3. Values derived from the left-hand side of Eq. (12) by using vapor pressures of CsBO_2 given by (a) Nakajima *et al.*,⁶⁾ (b) Biswas & Mukerji,¹⁸⁾ (c) Cordfunke *et al.*¹⁹⁾

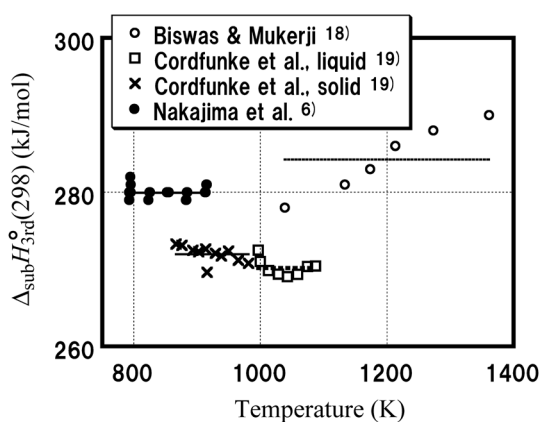


Fig. 4. Temperature dependence of the third law enthalpy of sublimation of CsBO_2 .^{6,18,19)}

at nine laboratories to establish vapor pressure standard reference materials.⁷⁾ These data sets of vapor pressures were used to confirm validity of the second and third law meth-

Table 2. The second-law and third-law sublimation enthalpies of Ag.

Lab. No.	$\Delta_{\text{sub}}H_{2\text{nd}}^{\circ}(298)^*$ (kJ/mol)	$\Delta_{\text{sub}}H_{3\text{rd}}^{\circ}(298)^*$ (kJ/mol)	Diff. (%)	$\Delta_{\text{sub}}H_{2\text{nd}}^{\circ}(298)^{\dagger}$ (kJ/mol)	$\Delta_{\text{sub}}H_{3\text{rd}}^{\circ}(298)^{\dagger}$ (kJ/mol)	Diff. (%)
1	287.3	285.1	0.8	287.7	284.9	1.0
2	290.4	281.5	3.2	290.4	281.4	3.2
3	304.0	284.6	6.8	304.4	284.3	7.1
4	293.6	286.6	2.4	293.9	286.3	2.7
5	286.2	285.7	0.2	286.6	285.5	0.4
6	287.4	285.2	0.8	287.8	285.0	1.0
7	290.2	283.2	2.4	290.2	283.1	2.5
8	250.4	285.5	12.3	250.8	285.4	12.1
9	320.4	279.5	14.6	320.8	279.3	14.9
Ave. [†]	289.2	284.6	1.6	289.4	284.3	1.8

*Gibbs energy functions of Ag(s, l) and Ag(g) given by Cox *et al.*²⁰⁾

[†]Gibbs energy functions of Ag(s, l) and Ag(g) given by Barin.²¹⁾

[†]The second law result from laboratory No. 3 and the second and third law results from laboratories Nos. 8 and 9 are discarded.

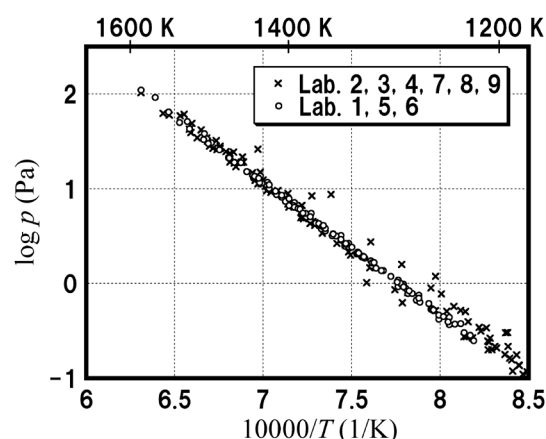


Fig. 5. Temperature dependence of vapor pressures of silver.⁷⁾

ods. Gibbs energy functions of silver given by Cox *et al.*²⁰⁾ and Barin²¹⁾ were used to identify their effect on the difference between the second and third law values. The examination results are summarized in Table 2. As shown in this table, the differences between the second and third law values using Gibbs energy functions given by Cox *et al.* are smaller than those in the case of Barin. Then, Cox *et al.*'s data is considered to be more reasonable. Further, it is shown that all the third law values of Lab. Nos. 1, 5 and 6, which have no greater than 1% of the difference between the second and third law values, lie between 285 and 286 kJ/mol and are close to the average third law value of 284.6 kJ/mol. Figure 5 shows all the data points of vapor pressures obtained from Lab. Nos. 1, 5 and 6 lie within the scatter of the data points obtained from other laboratories. Thus, examination of such a difference can lead to find more accurate data set of vapor pressures.

For comparison among 9 laboratories, the values derived from the left-hand side of Eq. (12) and from the right-hand side of Eq. (1) using Cox *et al.*'s Gibbs energy functions are plotted in Figs. 6 and 7, respectively. These figures show that, as the difference between the second and third law values becomes smaller, the intercept of the regression line approaches the origin and the temperature dependence of the third law values becomes weaker. Further, these plots can help us to easily find wrong data points which are largely deviated from the regression line or the average value. Therefore such a thermodynamic method including not only examination of the difference between the second and third

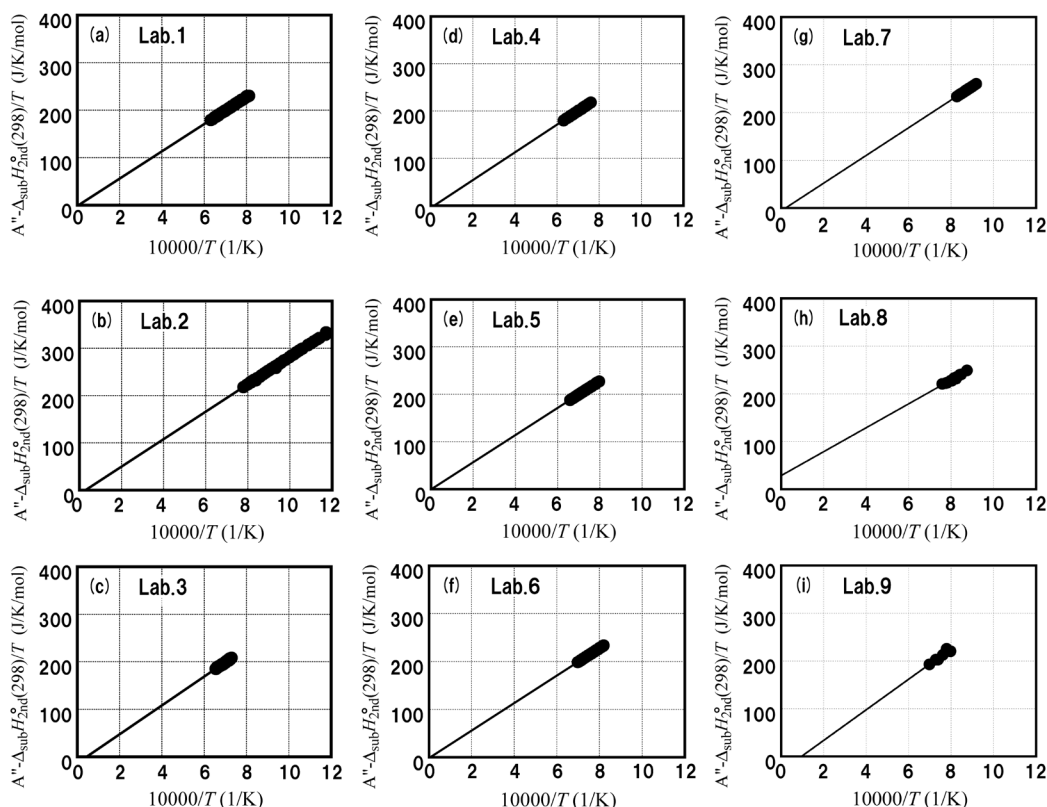


Fig. 6. Values derived from the left-hand side of Eq. (12) by using vapor pressures of silver⁷⁾ given by (a) laboratory No. 1, (b) laboratory No. 2, (c) laboratory No. 3, (d) laboratory No. 4, (e) laboratory No. 5, (f) laboratory No. 6, (g) laboratory No. 7, (h) laboratory No. 8, (i) laboratory No. 9.

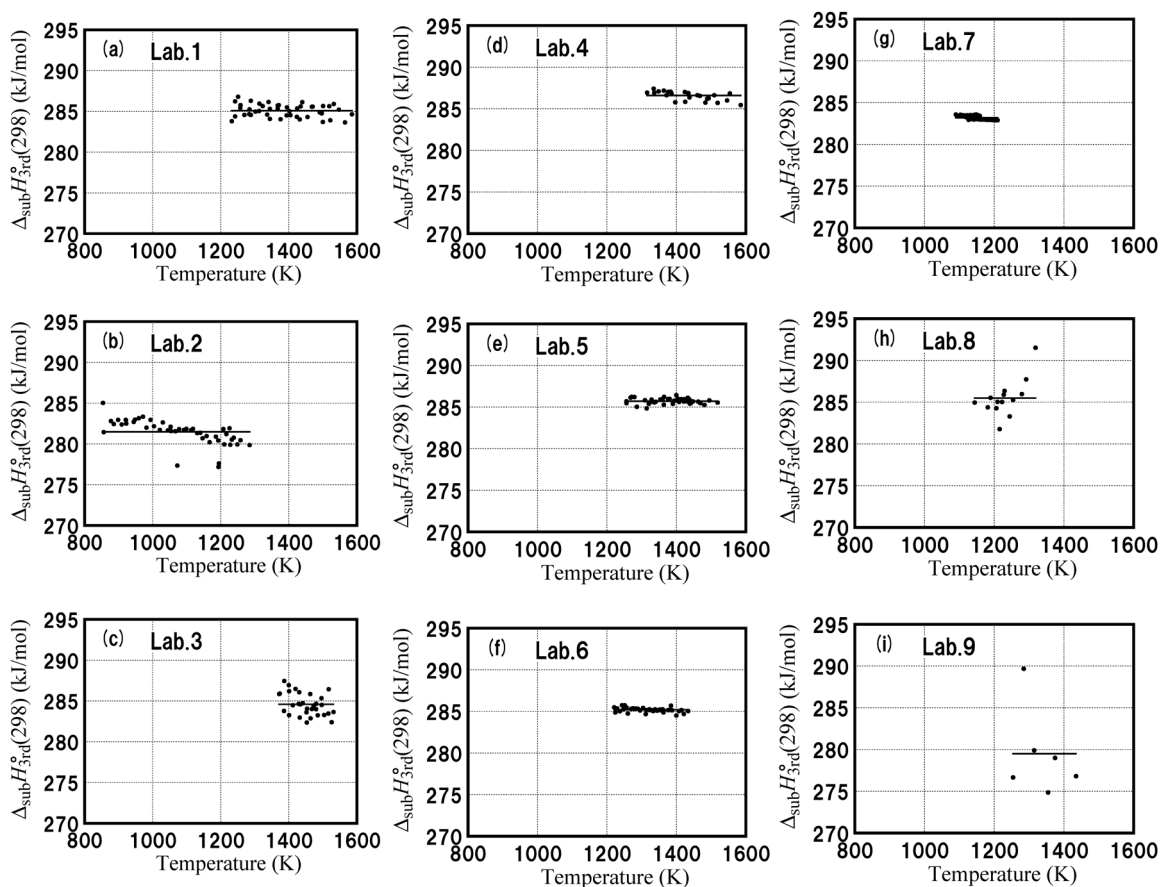


Fig. 7. Temperature dependence of the third law sublimation enthalpy of silver⁷⁾ given by (a) laboratory No. 1, (b) laboratory No. 2, (c) laboratory No. 3, (d) laboratory No. 4, (e) laboratory No. 5, (f) laboratory No. 6, (g) laboratory No. 7, (h) laboratory No. 8, (i) laboratory No. 9.

law values but also the plots such as shown in Figs. 6 and 7 can contribute to improvement of accuracy of vapor pressure measurement.

CONCLUSION

In this study, a new data analysis method is proposed to improve accuracy of vapor pressure measurements. The key of the new method is to use Eq. (12), which is derived from the van't Hoff equation based on the second law of thermodynamics. The proposed method using the second and third law methods is found to be helpful to determine an optimal data set of vapor pressures when it is hard to know only measured data such as time dependence of ion currents in KEMS measurement. Further, examination results for vapor pressures of silver indicate that a smaller difference between the second and third law values can lead to find more accurate vapor pressure data. Therefore, such a thermodynamic method can contribute to improve accuracy of vapor pressure measurement.

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REFERENCES

- 1) R. Hultgren, R. L. Orr, P. D. Anderson, K. K. Kelley. Selected Values of Thermodynamic Properties of Metals and Alloys, p. V, John Wiley, New York, 1963.
- 2) R. C. Paule, J. Mandel. Analysis of interlaboratory measurements on the vapor pressure of cadmium and silver. *Pure Appl. Chem.* 31: 395–431, 1972.
- 3) P. Chiotti, V. V. Akhachinskij, I. Ansara, M. H. Rand. The Chemical Thermodynamics of Actinide Elements and Compounds, Pt. 5, The Actinide Binary Alloys, IAEA, Vienna, 1981.
- 4) NIST-JANAF Thermochemical Tables, Fourth Edition, Monograph 9 (Pt. I and Pt. II) 1998, 1963 pp. by M. W. Chase, Jr.
- 5) J. Drowart, C. Chatillon, J. Hastie, D. Bonnell. High-temperature mass spectrometry: Instrumental techniques, ionization cross-sections, pressure measurements, and thermodynamic data. *Pure Appl. Chem.* 77: 683–737, 2005.
- 6) K. Nakajima, T. Takai, T. Furukawa, M. Osaka. Thermodynamic study of gaseous CsBO₂ by Knudsen effusion mass spectrometry. submitted to *J. Nucl. Mater.*
- 7) R. C. Paule, J. Mandel. Analysis of interlaboratory measurements on the vapor pressure of cadmium and silver. *Pure Appl. Chem.* 31: 395, 1972.
- 8) M. Miller, K. Armatys. Twenty years of Knudsen effusion mass spectrometry: Studies performed in the period 1990–2010. *Open Therm. J.* 7: 2–9, 2013.
- 9) R. T. Grimes. in *The Characterization of High Temperature Vapors* (Ed.: J. L. Margrave), Wiley, New York, 1967, p. 195.
- 10) J. Drowart, P. Goldfinger. Investigation of inorganic systems at high temperature by mass spectrometry. *Angew. Chem. Int. Ed. Engl.* 6: 581–596, 1967.
- 11) D. D. Cubicciotti. A new Σ plot treatment of equilibrium data and its application to the vaporization of bismuth chloride. *J. Phys. Chem.* 70: 2410–2413, 1966.
- 12) G. N. Lewis, M. Randall. Thermodynamics, 2nd Ed., revised by K. S. Pitzer and L. Brewer, McGraw-Hill, New York, 1961, p. 175.
- 13) R. A. Swalin. Thermodynamics of Solids, Wiley, New York, 1962, p. 51.
- 14) M. Asano, Y. Yasue, K. Kubo. Mass spectrometric study of ions formed from cesium metaborate vapor under electron impact. *J. Nucl. Sci. Technol.* 21: 614–624, 1984.
- 15) P. Clausing. The flow of highly rarefied gases through tubes of arbitrary length. *J. Vac. Sci. Technol.* 8: 636–646, 1971.
- 16) Yu. S. Ezhov, S. A. Komarov. Electron diffraction study of the molecular structure of cesium metaborate. *J. Struct. Chem.* 41: 55–60, 2000.
- 17) E. H. P. Cordfunke, R. J. M. Konings (Eds.). Thermochemical Data for Reactor Materials and Fission Products, North Holland, Amsterdam, 1990, p. 131.
- 18) S. R. Biswas, J. Mukerji. Vapor pressure of cesium metaborate. *J. Chem. Eng. Data* 16: 336–337, 1971.
- 19) E. H. P. Cordfunke, R. J. M. Konings, R. Saskia, M. Meyssen. Vapour pressures of some caesium compounds I. CsBO₂. *J. Chem. Thermodyn.* 23: 297–300, 1991.
- 20) J. D. Cox, D. D. Wagman, V. A. Medvedev. CODATA Key Values for Thermodynamics, Hemisphere Publishing Corp., New York, 1989.
- 21) I. Barin. Thermochemical Data of Pure Substances, 3rd Ed., VCH, Weinheim, 1995.